

KANNAN APPIYA SANTHARAM

Senior Lead Software Engineer (Frontend/Full Stack) | React · TypeScript · Node.js · AI-Native

Chennai, India

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Nationality: Indian | **Notice Period:** 60 Days

PROFESSIONAL SUMMARY

10.5+ years in enterprise SaaS, building across React, TypeScript, and Node.js. Currently leading an engineering team on a platform managing millions of customer endpoints, owning frontend architecture across a 12-package monorepo, design systems, build and runtime performance, alongside AI-native engineering with LLM orchestration and MCP servers. Seeking a Senior Lead / Staff Engineer or Engineering Manager role.

KEY ACHIEVEMENTS

- **AI test-authoring platform** that cut full product regression from 6 hours of manual testing to a 40-minute automated run.
- **Headless Base UI design system** with an AI agent that generates production components from a Figma link, taking page development from 2 days of AI-assisted work to a few hours.
- **96% faster dev builds**, 2 minutes to 5 seconds across 12 monorepo packages.
- **70% faster initial page load**, 5 seconds down to 1.5 seconds.
- **Hired 9 engineers** and mentored 10+ more across SuperOps and Freshworks, at 90% retention.

SKILLS

Frontend & Full Stack: React, Next.js, TypeScript, Node.js, Microservices, GraphQL & Apollo, REST API Design, JWT & OAuth 2.0, MySQL, DynamoDB, Python, Tailwind CSS, Playwright, Jest & Vitest

Platform & Performance: Monorepos & Module Federation, Rspack & Webpack 5, Micro-Frontends, Design Systems & UI SDKs, Core Web Vitals, Jenkins CI/CD, Docker, AWS (EC2, S3)

AI-Native Engineering: LLM Orchestration, Model Context Protocol (MCP) Servers, Agent Skills & Guardrails, RAG & Embeddings, Amazon Bedrock AgentCore, Claude Code

Leadership: Team Leadership (6 engineers), Hiring & Mentorship (10+ engineers), Solution Architecture & System Design, Stakeholder & Customer Management

WORK EXPERIENCE

SUPEROPS

Senior Lead Software Engineer

Chennai, India

Jul 2022 – Present

Progression: Senior Software Engineer to Lead Software Engineer to Senior Lead Software Engineer

- Hands-on delivery of enterprise features in React and TypeScript, including role-based access control (RBAC) and near-real-time operational dashboards tuned with custom cached Apollo hooks to balance data freshness against network load, in partnership with product and UI/UX teams.
- Lead and manage a frontend team of 6 engineers on a platform managing millions of customer endpoints, shipping weekly, owning sprint planning, architecture reviews, hiring and interviews, coaching and performance feedback, career development, and roadmap prioritisation with product managers.
- Reduced dev build time by 96%, from 2 minutes to 5 seconds, through a solo Webpack 5 to Rspack migration delivered in 3 weeks across 12 monorepo packages, reclaiming roughly 50 engineer-hours a month across a 15-engineer frontend org and accelerating time-to-market for every downstream feature.
- Reduced manual regression effort and release risk by architecting an AI test-authoring platform in React and Node.js that orchestrates LLM agents to generate, execute, and self-heal end-to-end test suites, streaming real-time results over HTTP Web Streams, cutting full product regression from 6 hours of manual testing to a 40-minute automated run in Jenkins CI/CD across a 300+ test-case Playwright suite.
- Improved developer productivity by integrating custom Claude Code skills and MCP servers into the production development workflow, with multi-tier write-scope guardrails that let engineers run autonomous agents without exposing core platform interfaces to risk.

- Built a Python MCP server exposing the frontend codebase as a queryable graph, packaged as a Docker image and hosted on AWS EC2 behind token-based authentication, now used by AI tooling across multiple engineering teams.
- Built a headless Base UI design system of 30+ reusable components now used across 40 to 50+ pages and sections, with customised design tokens and a Claude Code skill that generates production components from a Figma node link via MCP, reducing page development from 2 days of AI-assisted engineering to a few hours.
- Cut initial page load from 5 seconds to 1.5 seconds through route-level code splitting, React lazy loading (Suspense), and per-package bundle chunking, removing slowness-related support tickets, and eliminated 30,000+ lines of dead code and unused dependencies to reduce technical debt and the build and security surface.

FRESHWORKS

Senior Software Engineer (Customer-Facing Engineering)

Chennai, India

Jun 2018 – Jul 2022

Progression: Onboarding Engineer to Senior Software Engineer

- Acted as the technical point of contact for 100+ mid-market and enterprise accounts over a four-year tenure, partnering with Support, Customer Success, and Sales to turn complex account requirements into solutions that unblocked onboarding and renewals, recognised with the Customer Champion Award.
- Reduced support handling effort by building a Customer 360 dashboard consolidating support, sales, and account data into a single operational view.
- Built tier-1 SaaS integrations over REST API services, the Jira integration adopted by 4,000+ Freshworks customers, alongside Salesforce, Zendesk, and ServiceNow for 100+ mid-market and enterprise customers, expanding platform fit in enterprise deals.
- Shipped custom features and public apps across 4 Freshworks products, filling product gaps for enterprise accounts and reaching 10,000+ customers through the marketplace, with React on the front end and JWT for app authentication and OAuth 2.0 where the third-party API supported it.
- Built the Node.js microservices behind customer-specific features, deployed on AWS EC2 with S3, DynamoDB and MySQL as the data layer, so each customer's logic ran isolated from the core product.
- Mentored 6 to 7 junior and onboarding engineers on React, REST API development, and UI architecture through structured code and design reviews, accelerating team onboarding.

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Software Developer (acquired by Freshworks)

Chennai & Bengaluru, India

Mar 2016 – Jun 2018

Progression: Web Application Developer & Designer to Software Developer

- Built an admin dashboard that dynamically priced Stripe plans across feature selection, storage tiers, and bandwidth limits, increasing self-serve revenue.
- Built video player customisation from scratch with configurable controls, annotations, and call-to-action overlays, increasing viewer conversion.
- Built responsive, high-converting company and product landing pages.
- Integrated Freshservice with DocuSign over REST API services, unlocking a \$500K ARR enterprise deal, and joined Freshworks with the team on acquisition.

EDUCATION

Bachelor of Engineering (B.E.), Computer Science

2011 – 2015

K.L.N. College of Information Technology (Affiliated with Anna University), Tamil Nadu, India

LANGUAGES

English: Fluent (Professional Working Proficiency) | **Tamil:** Native | **Hindi:** Intermediate